

Nonparametric Density & Regression Estimation

Applied on MSE Returns

Riste Petrov
riste@uni-sofia.bg

AEEM - Sofia University St. Kliment Ohridski

March 1, 2026

SOFIA UNIVERSITY
ST. KLIMENT OHRIDSKI



Contents

1 Introduction

- Approach
- Theory

2 Empirical Approach

- Bandwidth calculation
- Kernel Density Estimation
- Local Linear Regressions
- Quantile Regressions
- Non-linear Quantile Regression vs Quantile Regression Estimation
- MISE analysis

3 Thank you for your attention!

- Q&A

Bibliography Basis

Theory

The main article used for these materials and research was **Nonparametric Density and Regression Estimation** (2001) by John DiNardo and Justin L. Tobias.

Key limitations and findings of the technique include:

- Limitations in selecting appropriate bandwidth related to over-smoothing and under-smoothing.
- Calculation of standard errors or other measures of model accuracy.

Similarity to Value at Risk (VaR)

Theory

What actually VaR is, and how it is similar?

- Value at Risk (VaR) is a financial metric used to assess the potential loss in value of an asset or portfolio over a defined period for a given confidence interval. The concept is typically applied in risk management to quantify the level of financial risk within a firm or investment portfolio.
- One tailed test, usually set at 0.05 or 0.01 percentile.

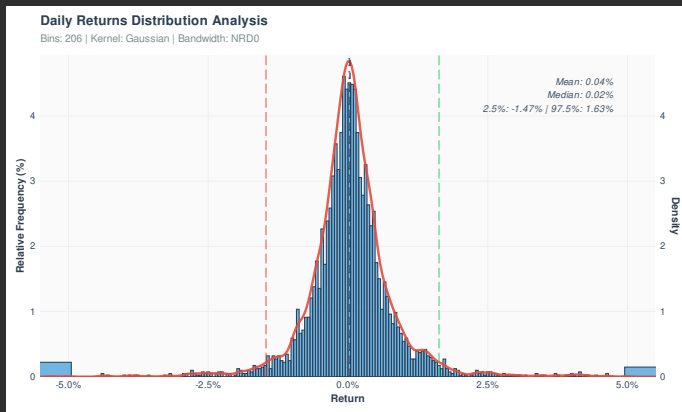
Theorem 1 (Value at Risk)

$$VaR_{\alpha} = -\inf\{x \in \mathbb{R} : P(X \leq x) \geq \alpha\}$$

Kernel Density Estimation (KDE) similarity

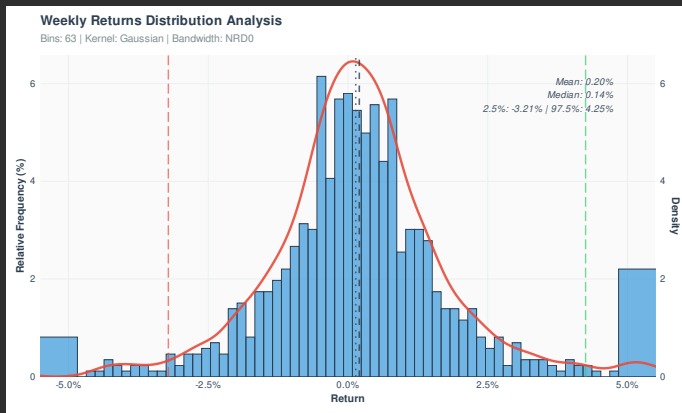
Practical Approach

Using the daily, weekly, monthly, yearly returns, we could better visualize the possible returns using a KDE than with just using the VaR formula.



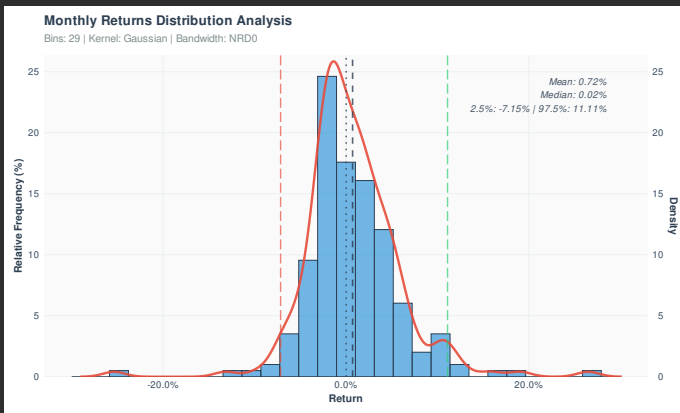
Kernel Density Estimation (KDE) similarity

Practical Approach



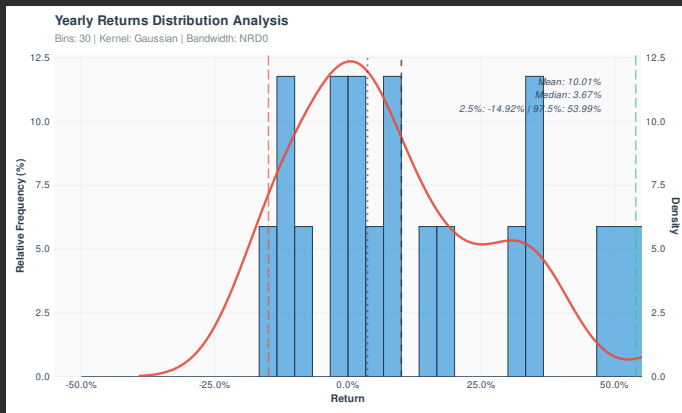
Kernel Density Estimation (KDE) similarity

Practical Approach



Kernel Density Estimation (KDE) similarity

Practical Approach



Kernel Density Estimation (KDE)

Theory

Kernel Density Estimation (KDE) is a non-parametric way to estimate the probability density function of a random variable. It smooths the observed data points to create a continuous probability density function.

$$\hat{f}(u) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{u - u_i}{h}\right)$$

Bandwidth Selection

Theory

Three most used bandwidths are:

- Normal Reference Distribution (nrd / 0)

$$h_{nrd} = 0.9sn^{-\frac{1}{5}}, \text{ when } s=0, \text{ then } h=1 \text{ in nrd0}$$

- Sheather - Jones (SJ)

$$h_{SJ} = s \left(\frac{4}{3n} \right)^{\frac{1}{5}}$$

- Unbiased Cross Validation (ucv)

$$h_{ucv} = \arg \min_h \left(\frac{1}{n} \sum_{i=1}^n K_h(x_i - u) f(u) du - \frac{1}{n} \sum_{i=1}^n K_h(x_i) \right)$$

Kernel Selection

Theory

- Rectangular - $\mathbb{I}$ - flat influence over the bandwidth. $K(u) = \frac{1}{2}$
- Epanechnikov - $\mathbb{I}$ - Best performing for MISE. $K(u) = \frac{3}{4}(1 - u^2)$
- Biweight - $\mathbb{I}$ - smoother estimate. $K(u) = \frac{15}{16}(1 - u^2)^2$
- Triangular - $\mathbb{I}$ - Balance between ease of computation and smoothness

$$K(u) = (1 - |u|)$$

- Gaussian

$$K(u) = \frac{1}{\sqrt{2\pi}\sigma'} e^{-\frac{u^2}{2}}$$

Local Linear Regression

Theory

Local Linear Regression is a non-parametric regression method that fits a linear function to a local subset of the data around each point of interest. This approach allows for flexibility in estimating complex relationships in the data.

$$\hat{y}(x_0) = \frac{\sum_{i=1}^n K\left(\frac{x_i - x_0}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x_i - x_0}{h}\right)}$$

Quintile Regression

Theory

Quintile Regression is a statistical technique that estimates the conditional median or other quantiles of the response variable, allowing for a comprehensive analysis of the relationship between variables beyond just the mean.

$$Q_y(\tau|X) = \frac{\sum_{i=1}^n K\left(\frac{y_i - Q_y(\tau|X)}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{y_i - Q_y(\tau|X)}{h}\right)}$$

Mean Integrated Square Error

Theory

The Mean Integrated Square Error (MISE) is used to evaluate the performance of density estimators. It takes into account both bias and variance, measuring how well the estimated density approximates the true density.

$$\text{MISE} = \int \left(\hat{f}(x) - f(x) \right)^2 dx$$

$$\text{MISE} = \text{Bias}^2 + \text{Variance}$$

Bandwidth calculation

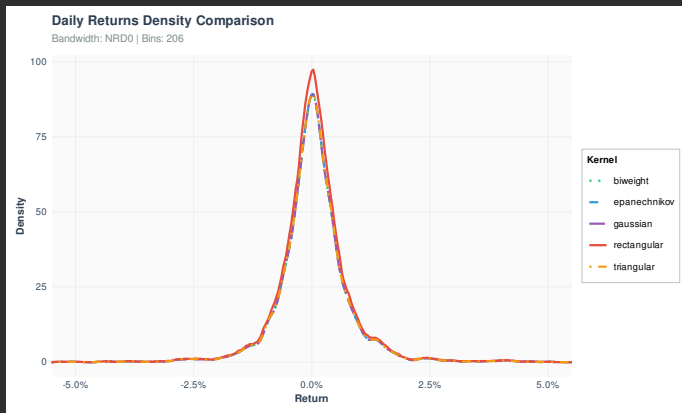
Across periods

Period	n_points	Data Range	nrd0	SJ	UCV
Daily	4057	6031	297.83	166.18	NA
Weekly	862	6027	405.89	295.50	NA
Monthly	199	6016	547.28	508.03	562.38
Yearly	17	5833	941.36	1271.17	NA

Period	n_points	Data Range	nrd0_pct_of_range	SJ_pct_of_range	UCV_pct_of_range
Daily	4057	6031	4.94	2.76	NA
Weekly	862	6027	6.73	4.90	NA
Monthly	199	6016	9.10	8.44	9.35
Yearly	17	5833	16.14	21.79	NA

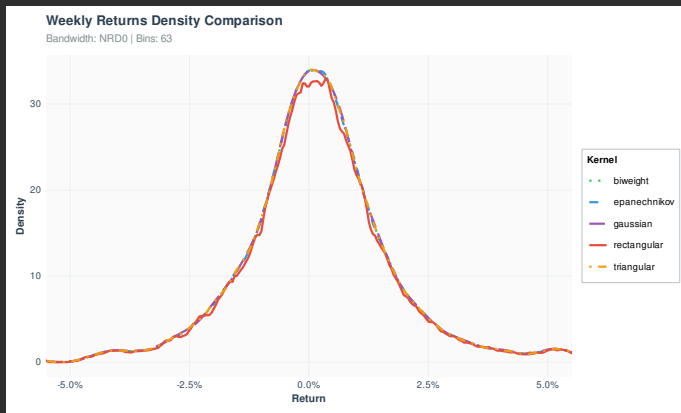
KDE Overlay

Bandwidth



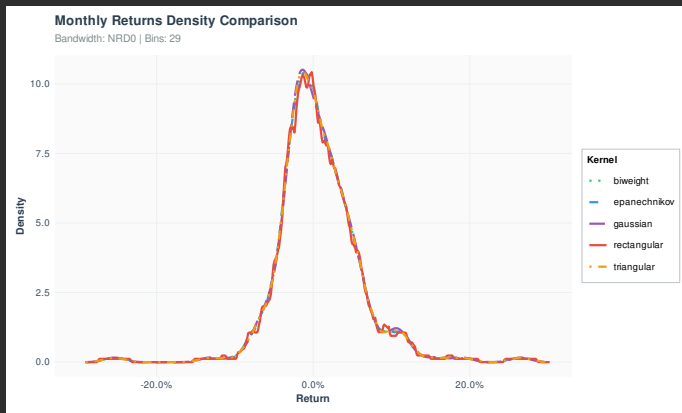
KDE Overlay

Bandwidth



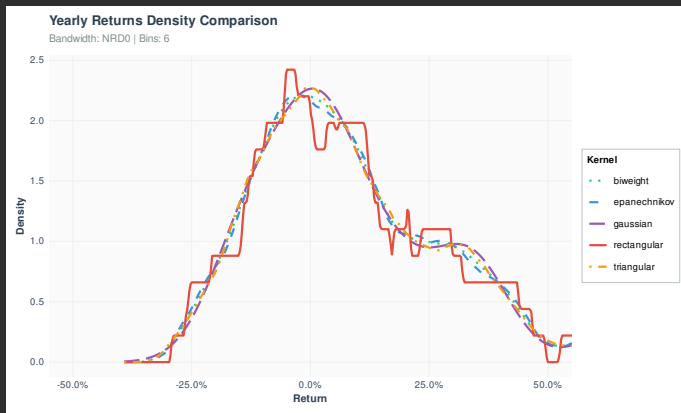
KDE Overlay

Bandwidth



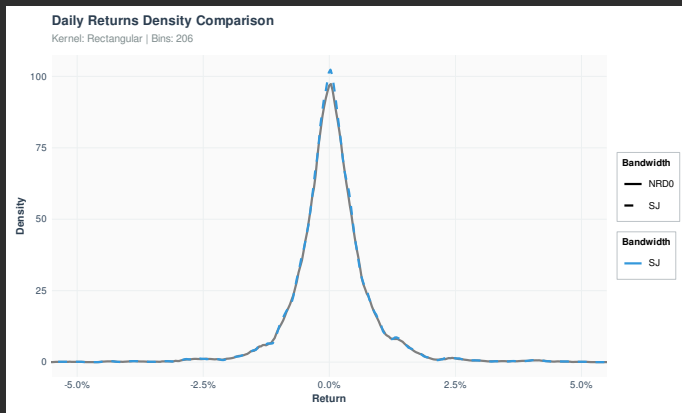
KDE Overlay

Bandwidth



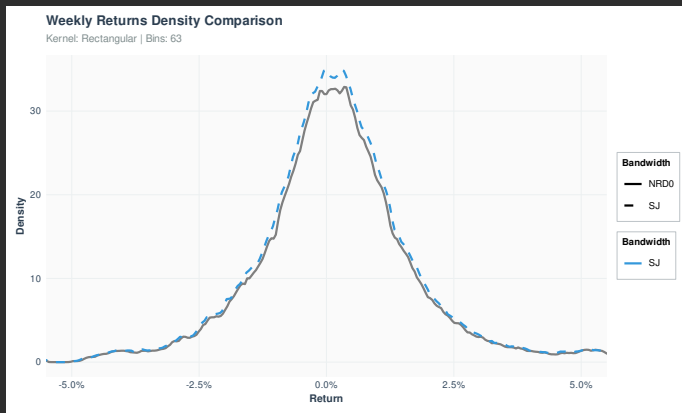
KDE Overlay

Kernel



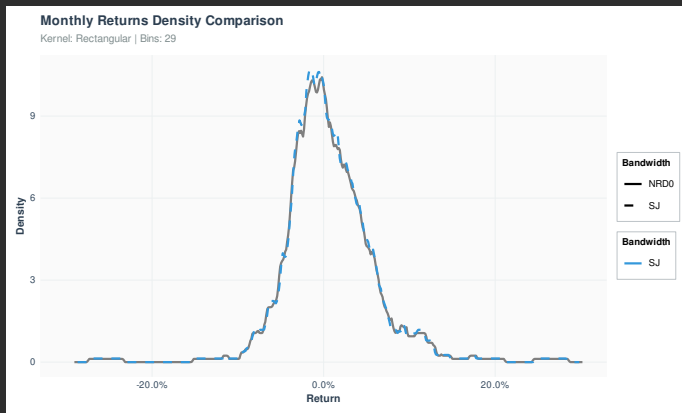
KDE Overlay

Kernel



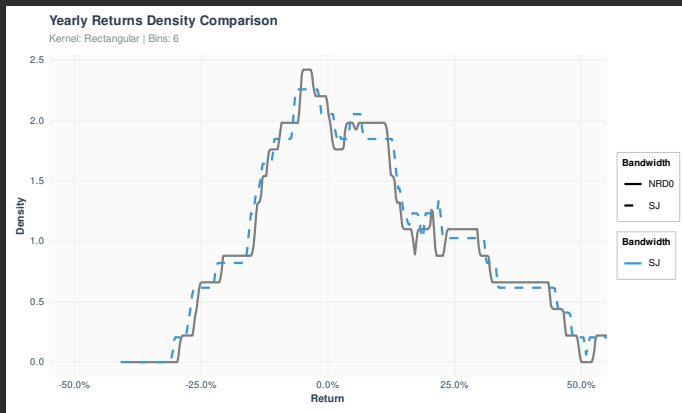
KDE Overlay

Kernel



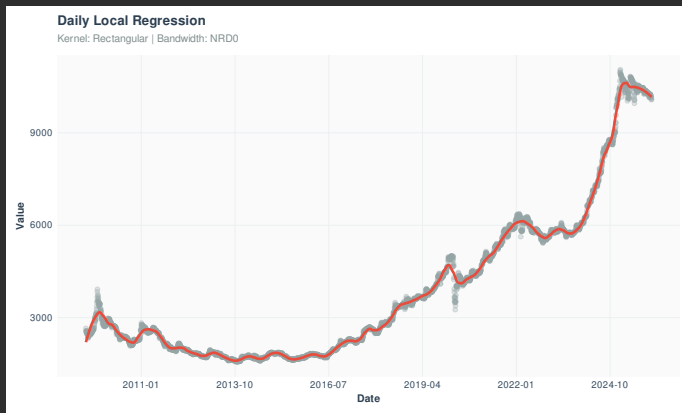
KDE Overlay

Kernel



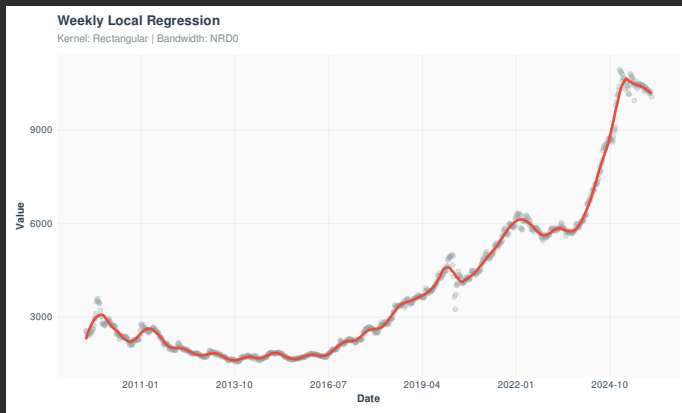
Local Linear Regressions

Daily



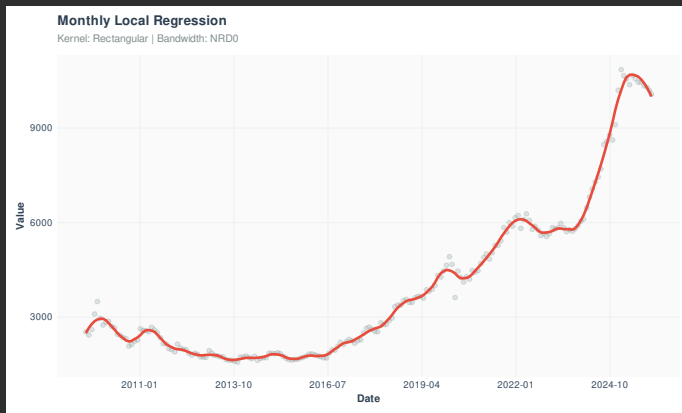
Local Linear Regressions

Weekly



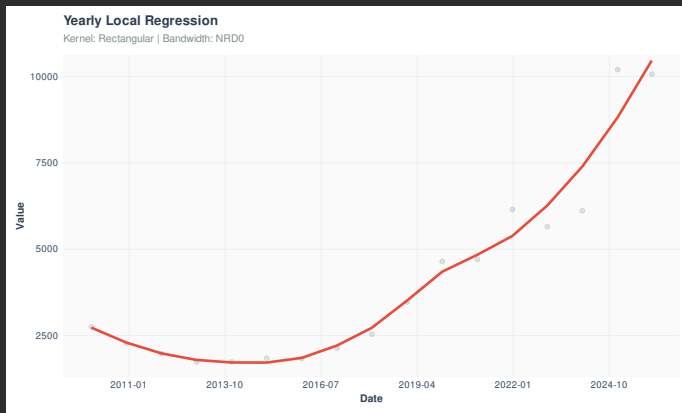
Local Linear Regressions

Monthly



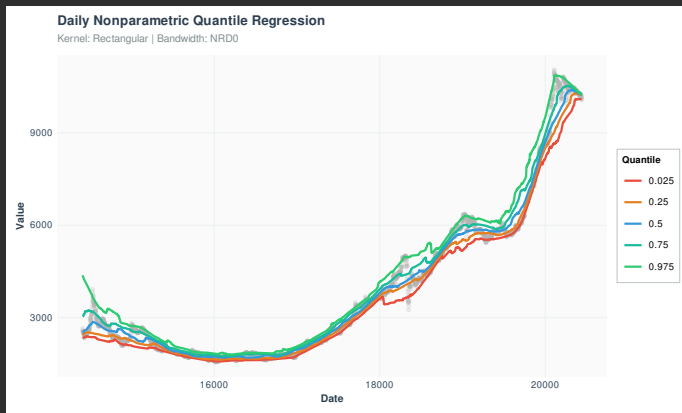
Local Linear Regressions

Yearly



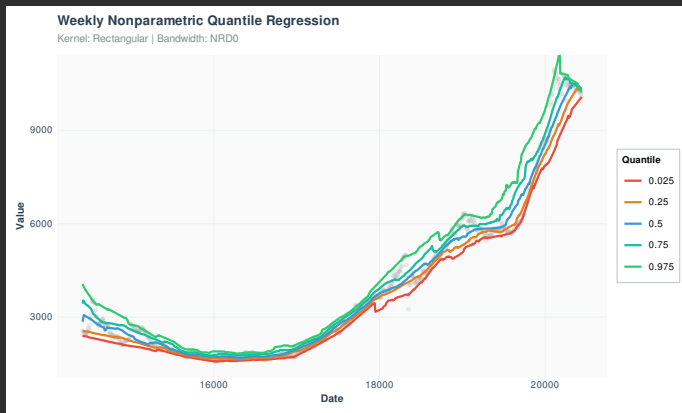
Quantile Regressions

Daily



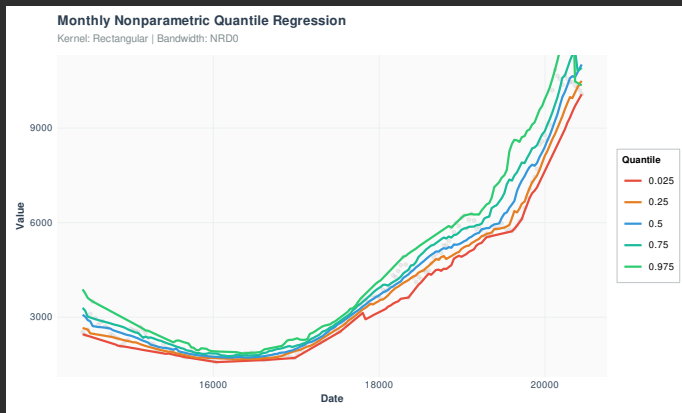
Quantile Regressions

Weekly



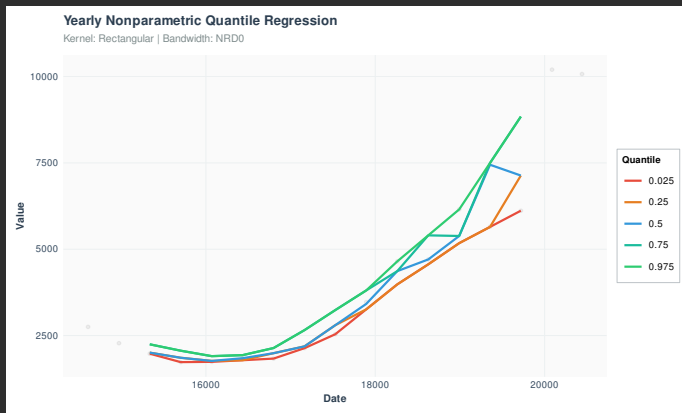
Quantile Regressions

Monthly



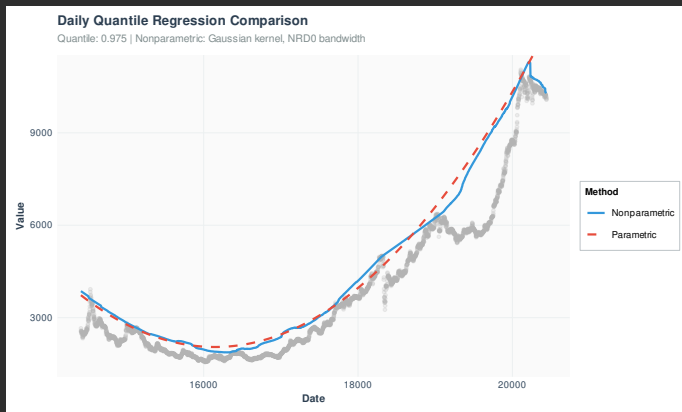
Quantile Regressions

Yearly



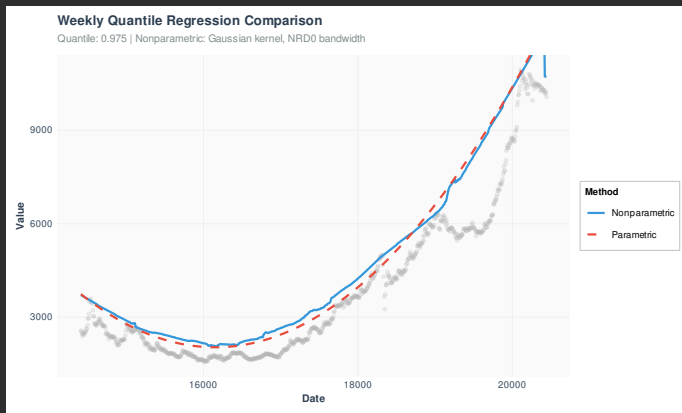
Non-linear Comparison

Daily



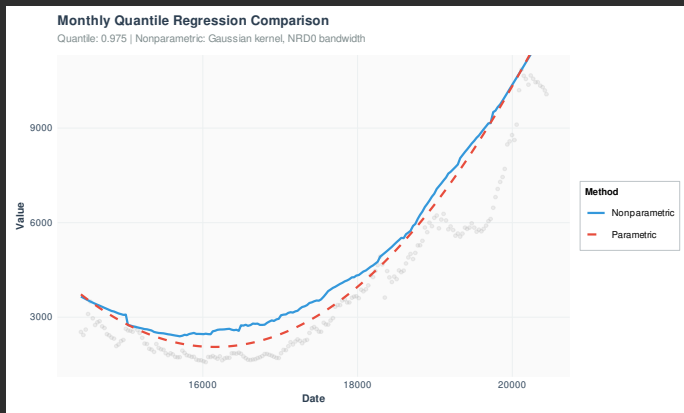
Non-linear Comparison

Weekly



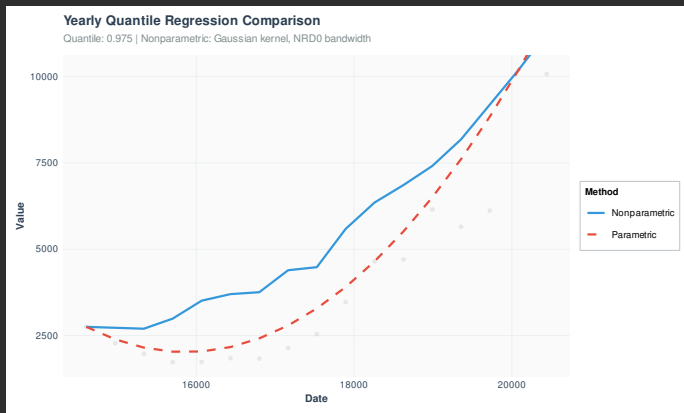
Non-linear Comparison

Monthly



Non-linear Comparison

Yearly



Mean Integrated Square Error

Best Model Estimates - Daily and Weekly

Period	Config	MISE
Daily	gaussian_nrd0_0.975	97429.72
Daily	gaussian_nrd0_0.5	122200.11
Daily	gaussian_nrd0_0.75	122700.55
Weekly	gaussian_nrd0_0.975	43577.16
Weekly	gaussian_nrd0_0.75	46878.43
Weekly	gaussian_nrd0_0.5	93805.95

Mean Integrated Square Error

Best Model Estimates - Monthly and Yearly

Period	Config	MISE
Monthly	gaussian_SJ_0.75	27661.85
Monthly	gaussian_nrd0_0.75	28499.24
Monthly	gaussian_ucv_0.75	28850.62
Yearly	epanechnikov_nrd0_0.975	25895.93
Yearly	biweight_nrd0_0.975	25895.93
Yearly	rectangular_nrd0_0.975	25895.93

Thank you for your attention

Q & A !